

13.3 Rayleigh Quotient



The core skills you'll master in this section:

- ▶ Rayleigh quotient: both the numerator and denominator are quadratic forms.
- ▶ The Rayleigh quotient's maximum and minimum values correspond to the matrix's largest and smallest eigenvalues, respectively.
- ▶ Understanding the two-variable Rayleigh quotient via polar coordinates.
- ▶ Observing how the Rayleigh quotient changes on the unit circle.
- ▶ Observing the Rayleigh quotient values corresponding to three-dimensional vectors on the unit sphere.
- ▶ Using a spherical latitude–longitude map to intuitively understand the three-variable Rayleigh quotient.

The Rayleigh quotient is a natural extension of the quadratic form — it is essentially a tool that expresses the relationship between a vector and a symmetric matrix in scalar form, and it is especially widely used in linear algebra and optimization problems.

To gain a deep understanding of the Rayleigh quotient's structure and meaning, we start from the simplest case, the two-variable Rayleigh quotient. With the help of the polar coordinate system and trigonometric functions, we can more clearly observe how the Rayleigh quotient changes on the two-dimensional unit circle. We then raise our perspective to three dimensions, using spherical coordinates to explore the geometric meaning of the Rayleigh quotient under a three-variable vector.

This section's content continuously weaves together and connects several familiar topics — including polar coordinates, trigonometric functions, the spherical coordinate system, quadratic-form structure, eigenvalue decomposition, and orthogonal transformations — to help you build a deep understanding of the Rayleigh quotient across different dimensions, and to link these different knowledge blocks into a network.

Definition

Given a real symmetric matrix $\mathbf{A}$, its Rayleigh quotient is defined as:

$$R(\mathbf{x}) = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$

where $\mathbf{x}$ is the independent-variable column vector, that is, $[\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_D]^T$.

⚠ Note that in the Rayleigh quotient's numerator and denominator, $\mathbf{x}^T$ and $\mathbf{x}$ must never be canceled out!

Since the equation above's denominator cannot be 0, $\mathbf{x}$ cannot be the zero vector $\mathbf{0}$, that is, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_D$ cannot all be 0 at once.

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As shown in Figure 1 (a), the Rayleigh quotient's numerator $\mathbf{x}^T \mathbf{A} \mathbf{x}$ is matrix $\mathbf{A}$'s quadratic form with respect to the column vector $\mathbf{x}$.

As shown in Figure 1 (b), the denominator $\mathbf{x}^T \mathbf{x}$ is essentially also a quadratic form, that is $\mathbf{x}^T \mathbf{I} \mathbf{x}$. $\mathbf{x}^T \mathbf{x}$ is also equivalent to the vector $\mathbf{x}$'s L^2 norm squared, that is $\|\mathbf{x}\|_2^2$, abbreviated as $\|\mathbf{x}\|^2$.

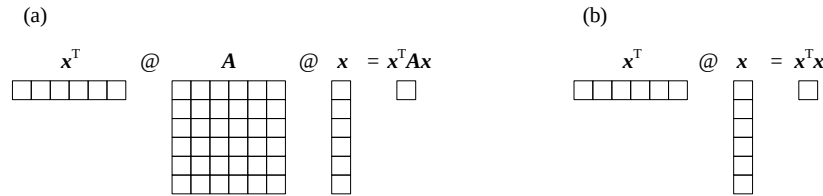


Figure 1. Matrix Operations for the Rayleigh Quotient's Numerator and Denominator

⚠ Note that a 1×1 matrix has only one element, so it behaves as a single numeric value. Mathematically, it is therefore equivalent to a scalar — only when both the Rayleigh quotient's numerator and denominator are treated as scalars can we form the fraction.

Maximum and Minimum Values

Let's start with the conclusion: the range of the Rayleigh quotient $R(\mathbf{x})$ is:

$$\lambda_{\min} \leq R(\mathbf{x}) \leq \lambda_{\max}$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are, respectively, matrix $\mathbf{A}$'s smallest and largest eigenvalues.

It's not hard to see that the Rayleigh quotient is, in essence, a multivariable function. For convenience of discussion, let's first look at an example of a two-variable Rayleigh quotient.

➡ To better understand how to compute a function's maximum and minimum values, we need some knowledge from optimization methods — this is a topic that "Math Made Easy," specifically Advanced Math Made Easy, will cover in depth.

Two-Variable Rayleigh Quotient: A Two-Variable Function

Given the following conditions

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

we can obtain the two-variable function corresponding to the Rayleigh quotient

$$R(x_1, x_2) = \frac{2x_1^2 + 2x_2^2 + 2x_1x_2}{x_1^2 + x_2^2}.$$

Since the denominator cannot be 0, this two-variable function's domain does not include the origin (0, 0). Simply put, a function's domain is the set of independent-variable values for which the function is defined.

For this two-variable function, it's easy to visualize its shape. Figure 2 shows this two-variable Rayleigh quotient's surface plot and its filled-contour plot. From the images, we can already see the function's maximum (local maximum) and minimum (local minimum) values.

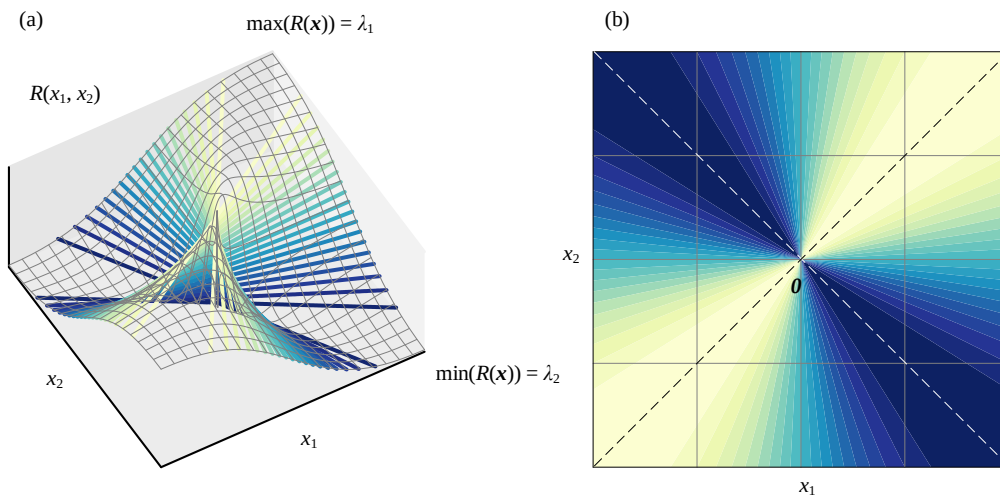
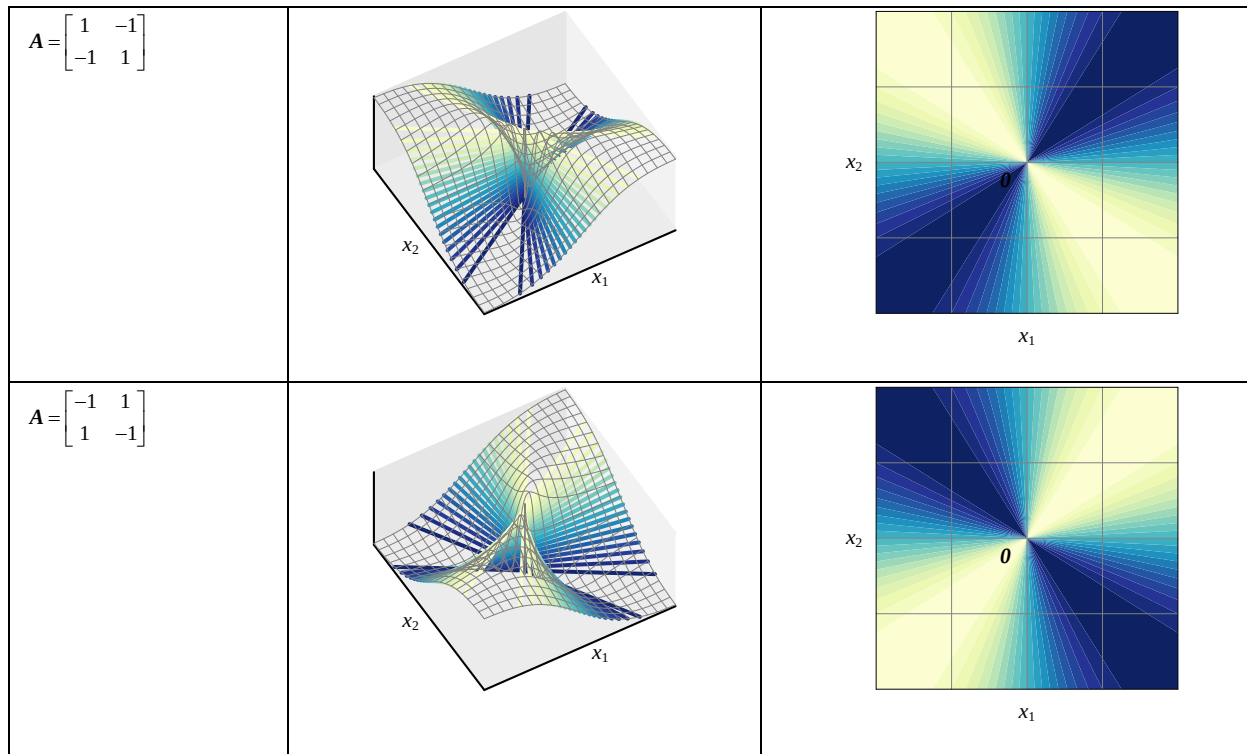


Figure 2. Surface Plot and Filled-Contour Plot of the Two-Variable Rayleigh Quotient $R(x)$

? Table 1 shows more examples of two-variable Rayleigh quotient surfaces and contour plots; go through each one and identify the Rayleigh quotient's maximum, minimum, and their specific locations.

Table 1. Two-Variable Rayleigh Quotient Surfaces and Contour Plots

Matrix A	Surface	Contour
$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$		
$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$		



Polar Coordinates and the Unit Circle

We're surprised to find that the two-variable Rayleigh quotient's contour plot radiates outward from the center, which suggests that the Rayleigh quotient's value seems unrelated to the vector's magnitude, and is instead directly related only to its direction. This immediately brings to mind the mathematical tool of polar coordinates.

To verify our guess, we introduce planar polar coordinates

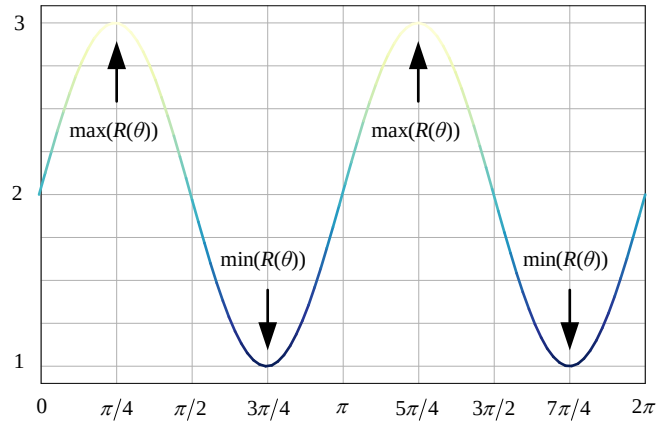
$$\begin{cases} x_1 = r \cos \theta \\ x_2 = r \sin \theta \end{cases}$$

Substituting them into the equation above's two-variable Rayleigh quotient, and simplifying, gives

$$\begin{aligned} R(\theta) &= \frac{2r^2 \cos^2 \theta + 2r^2 \sin^2 \theta + 2r^2 \cos \theta \sin \theta}{r^2} \\ &= 2 \cos^2 \theta + 2 \sin^2 \theta + 2 \cos \theta \sin \theta \\ &= 2 + \sin(2\theta) \end{aligned}$$

During the derivation, we canceled out r . This means the two-variable Rayleigh quotient depends only on the angle, confirming our earlier guess.

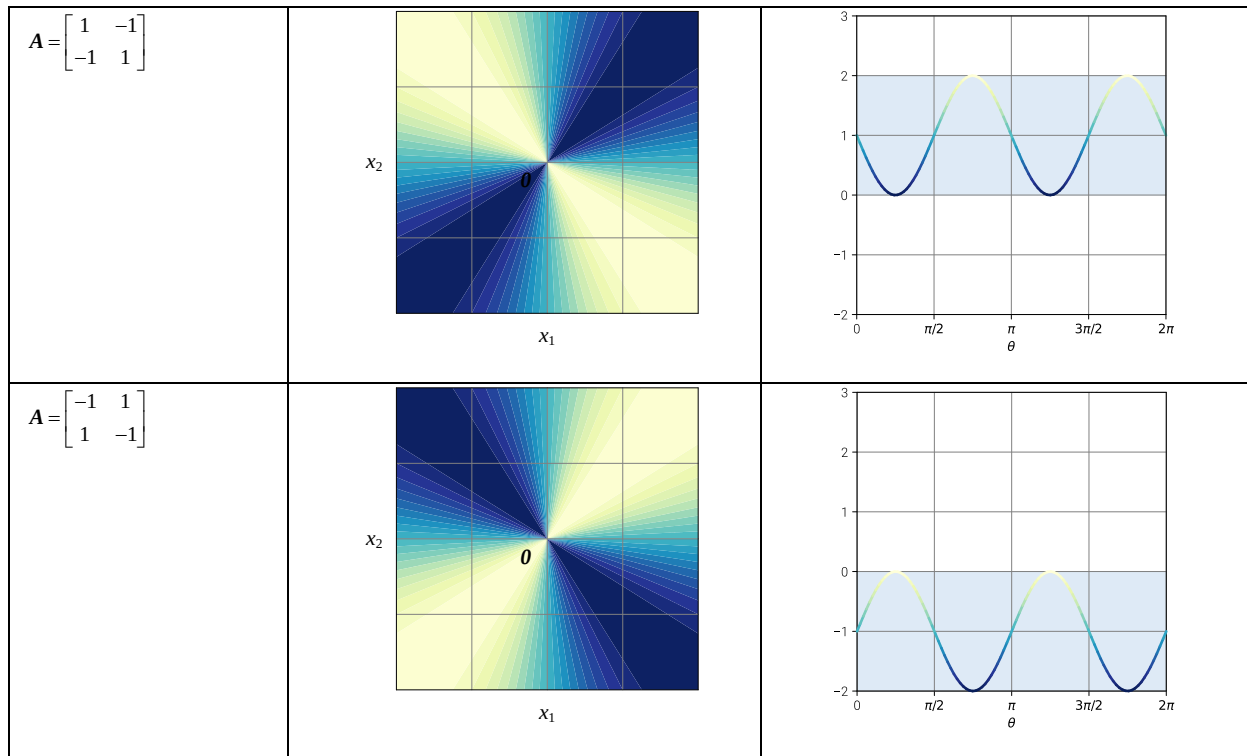
Figure 3 shows the trigonometric function corresponding to the equation above. We can easily find the Rayleigh quotient's maximum and minimum values, as well as the angles at which they occur.

Figure 3. The Trigonometric Function Corresponding to the Two-Variable Rayleigh Quotient $R(\mathbf{x})$

? Please go through each trigonometric function in Table 2 and find the two-variable Rayleigh quotient's maximum, minimum, and their corresponding specific locations.

Table 2. Two-Variable Rayleigh Quotient Contours and Trigonometric Functions

Matrix A	Contour	Trigonometric Function
$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$		
$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$		



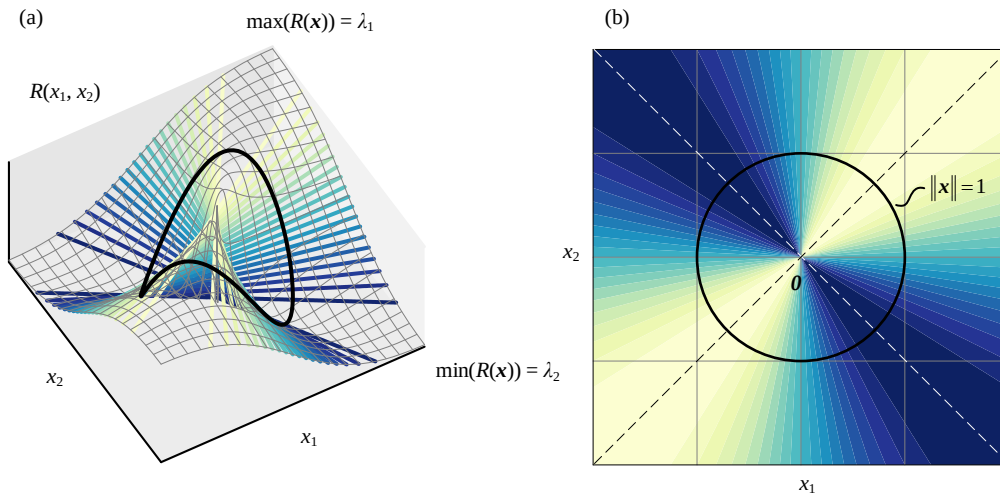
Viewing the Two-Variable Rayleigh Quotient on the Unit Circle

Precisely because the Rayleigh quotient's denominator can be written as $\|\mathbf{x}\|_2^2$, we can write the Rayleigh quotient as

$$R(\mathbf{x}) = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\|\mathbf{x}\|^2} = \left(\frac{\mathbf{x}}{\|\mathbf{x}\|} \right)^T \mathbf{A} \left(\frac{\mathbf{x}}{\|\mathbf{x}\|} \right) = \hat{\mathbf{x}}^T \mathbf{A} \hat{\mathbf{x}}$$

where $\hat{\mathbf{x}}$ is $\mathbf{x}$'s unit vector (direction vector).

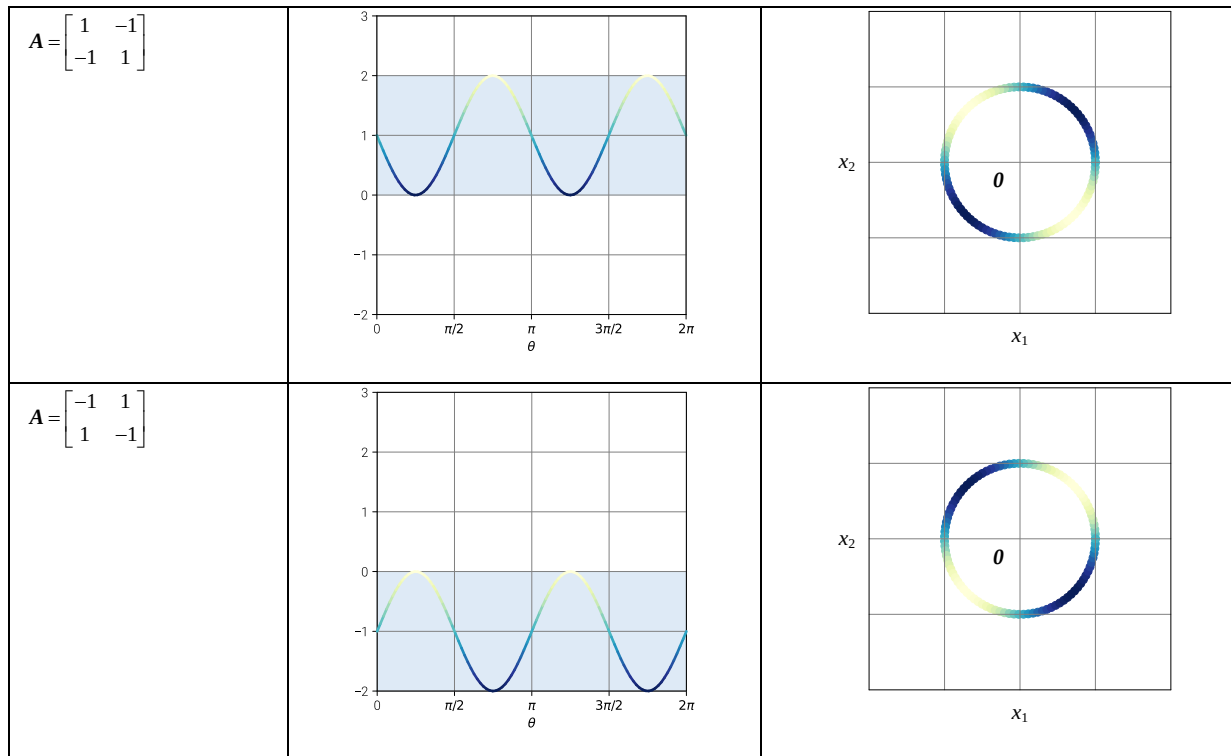
For the two-variable Rayleigh quotient in the equation above, the unit vector $\hat{\mathbf{x}}$'s endpoint always lies on the circumference of the unit circle. As shown in Figure 4, this means we can also observe the Rayleigh quotient on the unit circle.

Figure 4. Filled-Contour Plot of the Two-Variable Rayleigh Quotient $R(\mathbf{x})$, Observed on the Unit Circle

? Please analyze each example in Table 3.

Table 3. Viewing the Two-Variable Rayleigh Quotient on the Unit Circle

Matrix A	Trigonometric Function	Unit Circle
$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$		
$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$		



Eigenvalue Decomposition

For the matrix in the equation above, A 's eigenvalue decomposition (spectral decomposition) gives:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \underbrace{V}_{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}} \underbrace{\Lambda}_{\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}} \underbrace{V^T}_{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}$$

Writing V as column vectors v_1 and v_2 , as shown in Figure 5, v_1 and v_2 correspond, respectively, to the directions of the Rayleigh quotient's maximum value λ_1 and minimum value λ_2 .

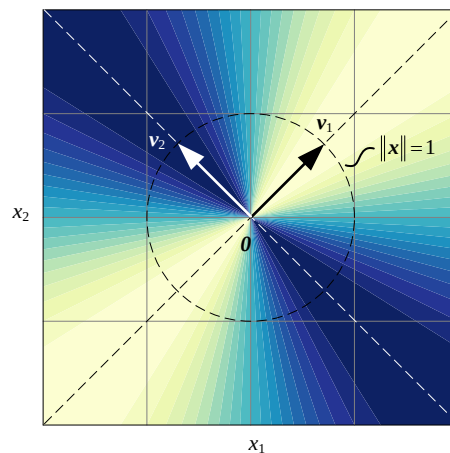


Figure 5. Directions Corresponding to the Two-Variable Rayleigh Quotient $R(x)$'s Maximum and Minimum Values

For example, in the direction of $\mathbf{v}_1$, the Rayleigh quotient attains its maximum value

$$\mathbf{v}_1^T \mathbf{A} \mathbf{v}_1 = \mathbf{v}_1^T \mathbf{V} \mathbf{A} \mathbf{V}^T \mathbf{v}_1 = \mathbf{v}_1^T \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} \mathbf{A} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \end{bmatrix} \mathbf{v}_1 = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 \end{bmatrix} \mathbf{A} \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 \\ \mathbf{v}_2^T \mathbf{v}_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \lambda_1$$

As mentioned earlier in this book, the equation above is a special form of the Rayleigh quotient.

As another example, in the direction of $\mathbf{v}_2$, the Rayleigh quotient attains its minimum value

$$\mathbf{v}_2^T \mathbf{A} \mathbf{v}_2 = \mathbf{v}_2^T \mathbf{V} \mathbf{A} \mathbf{V}^T \mathbf{v}_2 = \mathbf{v}_2^T \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} \mathbf{A} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \end{bmatrix} \mathbf{v}_2 = \begin{bmatrix} \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 \end{bmatrix} \mathbf{A} \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_2 \\ \mathbf{v}_2^T \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \lambda_2$$

Three-Variable Rayleigh Quotient

As introduced earlier in this book, if a two-dimensional unit vector's starting point is at the origin, its endpoint lies on the unit circle; similarly, a three-dimensional unit vector's endpoint lies on the unit sphere.

We can view the two-variable Rayleigh quotient on the unit circle; naturally, we can likewise observe the three-variable Rayleigh quotient on the surface of the unit sphere.

For example, given the following 3×3 symmetric matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0.5 & 1 \\ 0.5 & 2 & -0.2 \\ 1 & -0.2 & 1 \end{bmatrix}$$

the Rayleigh quotient corresponding to the symmetric matrix above is

$$f(x_1, x_2, x_3) = \frac{x_1^2 + 2x_2^2 + x_3^2 + x_1x_2 + 2x_1x_3 - 0.4x_2x_3}{x_1^2 + x_2^2 + x_3^2}$$

Figure 6 shows the above three-variable Rayleigh quotient viewed on the unit sphere.

Matrix $\mathbf{A}$'s three eigenvalues are 2.272, 1.844, and -0.117 (keeping only three decimal places). Therefore, from the equation above, the Rayleigh quotient's maximum value is 2.272, and its minimum value is -0.117.



Please compute, using the equation above, the eigenvectors of matrix $\mathbf{A}$.

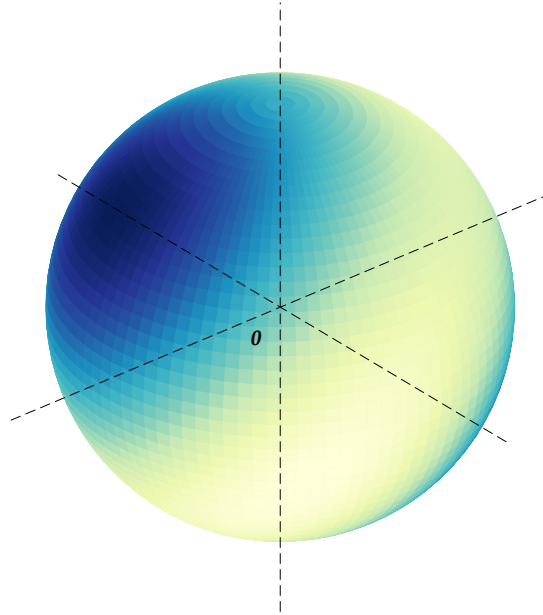


Figure 6. Observing the Three-Variable Rayleigh Quotient on the Unit Sphere

To more easily see the full picture of the three-variable Rayleigh quotient, imagine the sphere in Figure 6 as a globe — we can create a "Rayleigh quotient" world map, using latitude and longitude as coordinates to display the Rayleigh quotient.

As shown in Figure 7, θ is the angle between the line OP and the positive direction of the x_3 axis, called the polar angle (polar angle); the spherical coordinate θ ranges over $[0, \pi]$. This angle corresponds to the earth's "latitude angle".

The line OP 's projection onto the x_1x_2 plane is the line OH ; φ is the angle between OH and x_1 's positive-direction axis, called the azimuth angle (azimuth angle). The spherical coordinate φ ranges over $[0, 2\pi]$. This angle corresponds to the earth's "longitude angle".



Please review the spherical coordinates introduced in Chapter 1, Section 5 of this book.

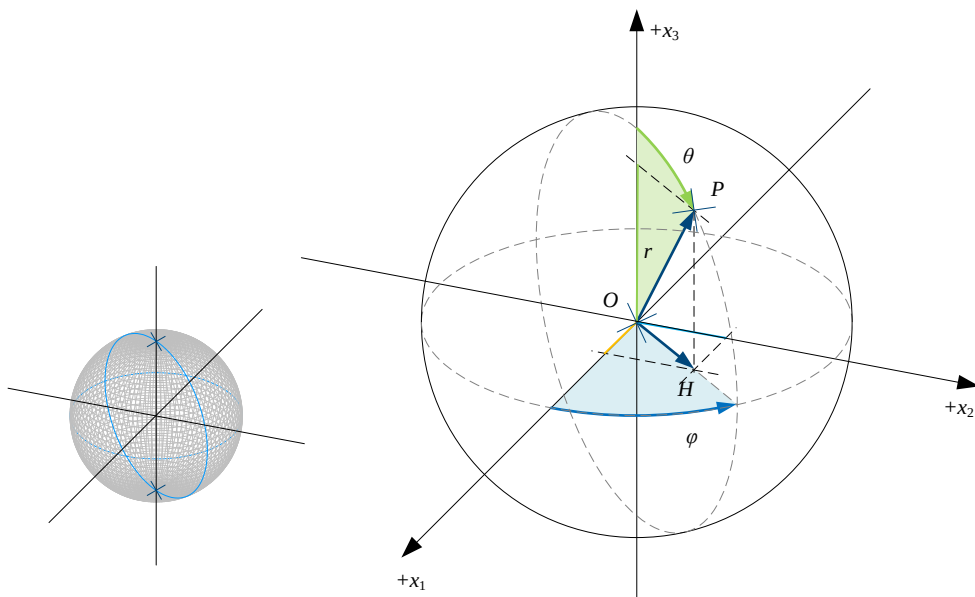


Figure 7. The Spherical Coordinate System (image from The Beauty of Visualization)

Figure 8 shows the three-variable Rayleigh quotient viewed from a "latitude-longitude" perspective.

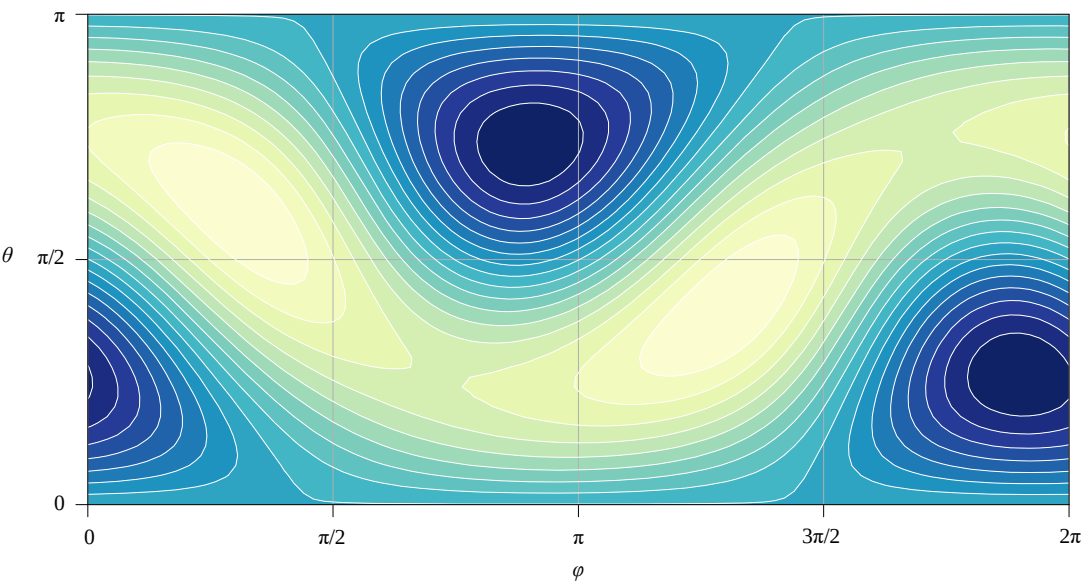
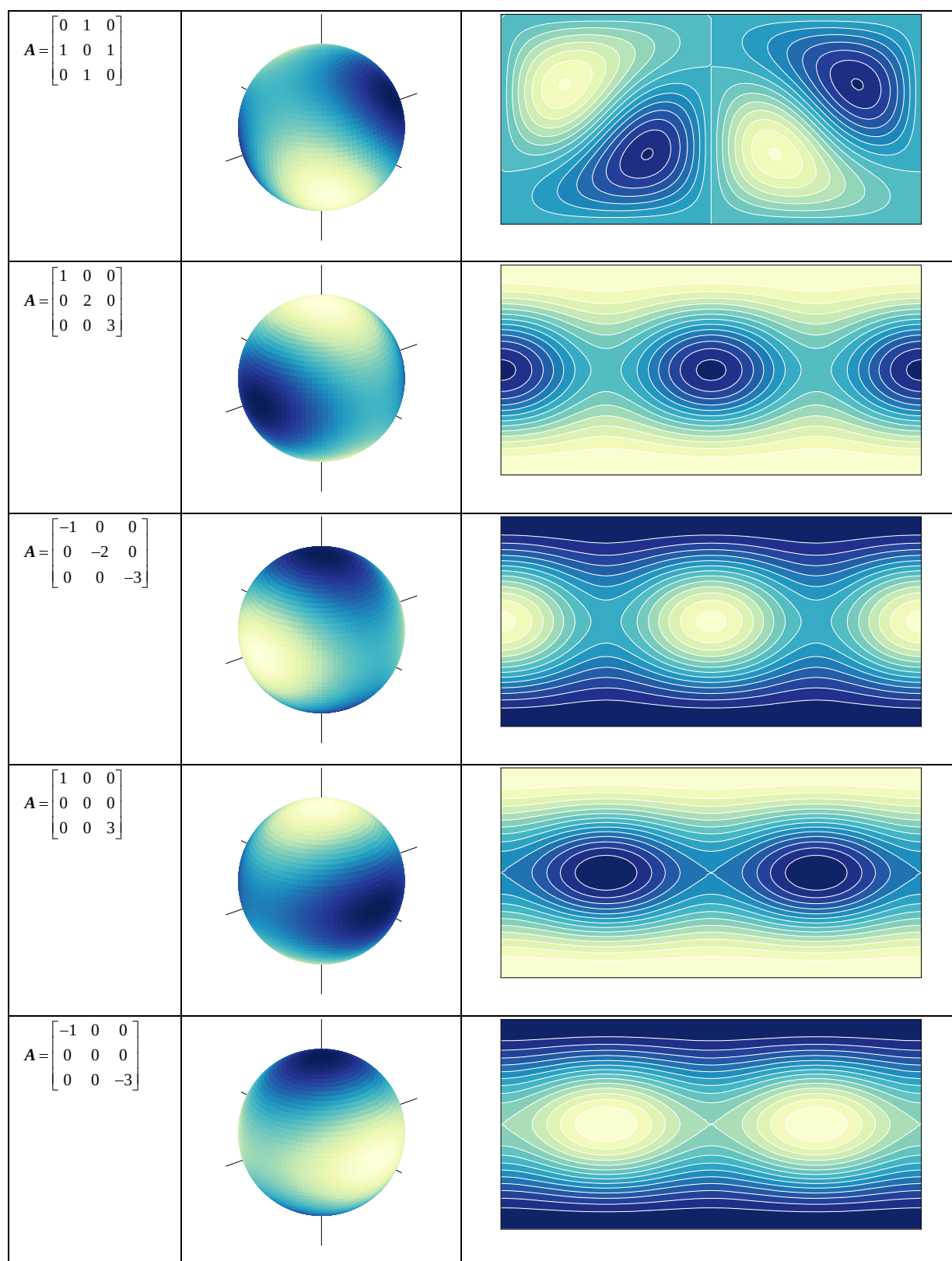


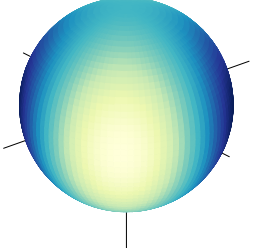
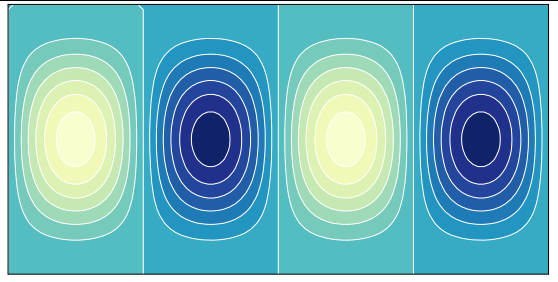
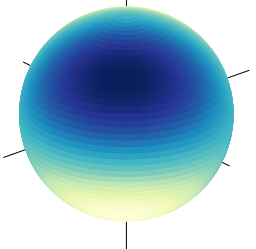
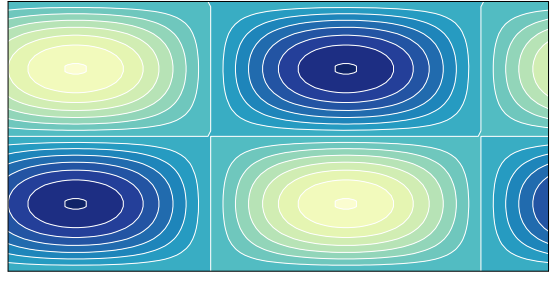
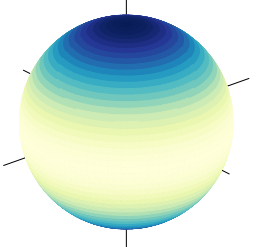
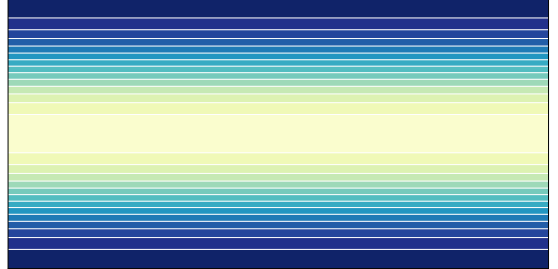
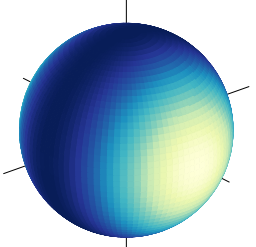
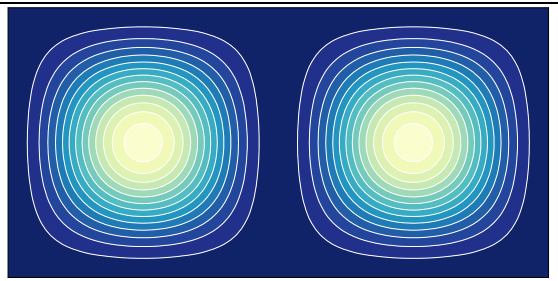
Figure 8. The Three-Variable Rayleigh Quotient Viewed from a "Latitude-Longitude" Perspective

? Please go through the three-variable Rayleigh quotients shown in Table 4 one by one, and identify the locations of the maximum and minimum values.

Table 4. Viewing the Three-Variable Rayleigh Quotient on the Unit Sphere

Matrix A	Unit Sphere View	Latitude–Longitude View
$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$		
$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$		



$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$		
$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$		
$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$		
$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$		



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. In Table 1, please compute the eigenvalue decomposition results – the eigenvalues and eigenvectors – of matrix A , and identify the corresponding Rayleigh quotient's maximum and minimum values.

Q2. Please compute each trigonometric function in Table 2, and identify the maximum and minimum values along with their corresponding angles.

Q3. In Table 3, please compute the Rayleigh quotient's maximum and minimum values for matrix A .

Q4. Please learn about the generalized Rayleigh quotient.

